GUILLAUME ST-ONGE

Professeur adjoint

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Science des réseaux | Processus stochastiques | Systèmes dynamiques | Inférence bayésienne | Contagion

POSTES ACADÉMIQUES

Professeur adjoint | École de science informatique et de génie électrique, Université d'Ottawa 2026–présent

Professeur-chercheur adjoint | Département de physique, Northeastern University 2024–2026

- Membre du corps professoral au Network Science Institute & Roux Institute
- Membre de EPISTORM—*Center for Advanced Epidemic Analytics and Predictive Modeling Technology*

Chercheur postdoctoral | Département de physique, Northeastern University 2022–2024

- Directeur de recherche : Alessandro Vespignani
- FRQNT : Boursier postdoctoral

FORMATION

Ph.D. en physique | Université Laval 2018–2022

- Directeurs : Antoine Allard et Laurent Hébert-Dufresne (co-directeur)
- Titre de thèse : *Contagion process on complex networks beyond pairwise interactions*

M.Sc. en physique | Université Laval 2015–2017

- Directeur : Louis J. Dubé
- Titre de thèse : *Propagation dynamics on random networks: characterization of the phase transition*

B.Sc. en physique | Université Laval 2012–2015

FINANCEMENT ET PRIX

Subventions

- Gates Foundation : *Advancing Global Pathogen Detection through Aircraft and WES* ($1 463 735 USD, co-PI) 2025

Recherche postdoctorale

- FRQNT : Bourse de recherche postdoctorale ($110 000) juin 2022–juin 2024

Recherche aux cycles supérieurs

- CRSNG : Bourse de doctorat – Alexander Graham Bell Canada ($105 000) janv. 2018–déc. 2020
- FRQNT : Bourse de doctorat* ($60 000) janv. 2018–déc. 2020
- CRSNG : Bourse de maîtrise – Alexander Graham Bell Canada ($17 500) sept. 2015–août 2016
- FRQNT : Bourse de maîtrise ($30 000) sept. 2015–août 2017
- Fondation Desjardins : Bourse de maîtrise* ($3 000) oct. 2015

*Accordée mais déclinée

Stages de recherche

- FRQNT : Programme de stages internationaux ($7 500) 2020
- CRSNG : Supplément pour études à l'étranger Michael-Smith ($6 000) 2019
- CRSNG : Bourse de recherche de 1er cycle ($4 500, accordée 3 fois) 2013, 2014, 2015

Prix et distinctions

- Tableau d'honneur de la Faculté des études supérieures | thèse de doctorat[†] 2022
- Meilleure présentation, Fourth Northeast Regional Conference on Complex Systems 2021
- Tableau d'honneur de la Faculté des études supérieures | thèse de maîtrise[†] 2017
- Concours d'expression scientifique Pierre Amiot (3e place), Université Laval 2017
- Médaille du gouverneur général (argent) pour excellence académique | B.Sc. 2016
- Gala du mérite étudiant de la faculté des sciences et de génie–mention de la direction, Université Laval 2015
- Pédagogue de l'année, Association des étudiants de physique, Université Laval 2014

ENSEIGNEMENT

- *Dynamical Processes in Complex Networks*, présentateur invité 2022–2026
  Titre de présentation : Branching processes and probability generating functions in network science
- Auxiliaire d'enseignement :
  - *Physique statistique* 2016–2018, 2020
  - *Physique numérique* 2016, 2018
  - *Physique mathématique III* 2014
  - *Physique mathématique I, II* 2013
- Livre en préparation : CoSMOS: Complex Systems Modeling Open Sourcebooks

PUBLICATIONS ET BREVETS

Articles publiés ou acceptés dans une revue avec un comité de lecture

25. *Evaluation of stochastic trajectory-based epidemic models using the energy score* 2026
C. Bay, K. Mu, **G. St-Onge**, M. Chinazzi, J. T. Davis, A. Vespignani
Epidemics. 56, 100929

24. *A vision for estimation of the instantaneous reproductive number* 2026
C. W. Milando, G. G. Vega Yon, K. Johnson, A. Urbinati, **G. St-Onge**, B. Klein, A. Cori, L. F. White, Rt Collabathon participants
Epidemics. 54, 100885

23. *Pandemic monitoring with global aircraft-based wastewater surveillance networks* 2025
G. St-Onge, J. T. Davis, L. Hébert-Dufresne, A. Allard, A. Urbinati, S. V. Scarpino, M. Chinazzi, A. Vespignani
Nature Medicine. 31, 788–796

22. *Characteristic scales and adaptation in higher-order contagions* 2025
G. Burgio, **G. St-Onge**, L. Hébert-Dufresne
Nature Communications. 16, 4589

[†]Meilleure note globale selon tous les membres du jury

21. *One pathogen does not an epidemic make: A review of interacting contagions, diseases, beliefs, and stories*
L. Hébert-Dufresne, Y.-Y. Ahn, V. Colizza, A. Allard, J. W. Crothers, P. Sheridan Dodds, M. Galesic, F. Ghanbarnejad, D. Gravel, R. A. Hammond, K. Lerman, J. Lovato, J. J. Openshaw, S. Redner, S. V. Scarpino, **G. St-Onge**, T. R. Tangherlini, J.-G. Young 2025
npj Complexity. 2, 26

20. *Ensemble^2: scenarios ensembling for communication and performance analysis*
C. Bay, **G. St-Onge**, J. T. Davis, M. Chinazzi, E. Howerton, J. Lessler, M. C. Runge, K. Shea, S. Truelove, C. Viboud, A. Vespignani 2024
Epidemics. 46, 100748

19. *Nonlinear bias toward complex contagion in uncertain transmission settings*
G. St-Onge, L. Hébert-Dufresne, A. Allard 2024
Proceedings of the National Academy of Sciences of the United States of America. 121, e2312202121

18. *Hierarchical team structure and multidimensional localization (or siloing) on networks*
L. Hébert-Dufresne, **G. St-Onge**, J. Meluso, J. Bagrow, A. Allard 2023
Journal of Physics: Complexity. 4, 035002

17. *Source-sink behavioural dynamics limit institutional evolution in a group-structured society*
L. Hébert-Dufresne, T. M. Waring, **G. St-Onge**, M. T. Niles, L. K. Corlew, M. P. Dube, S. J. Miller, N. J. Gotelli, B. J. McGill 2022
Royal Society Open Science. 9, 211743

16. *Influential groups for seeding and sustaining nonlinear contagion in heterogeneous hypergraphs*
G. St-Onge, I. Iacopini, V. Latora, A. Barrat, G. Petri, A. Allard, L. Hébert-Dufresne 2022
Communications Physics. 5, 25

15. *Universal Nonlinear Infection Kernel from Heterogeneous Exposure on Higher-Order Networks*
G. St-Onge, H. Sun, A. Allard, L. Hébert-Dufresne, G. Bianconi 2021
Physical Review Letters. 127, 158301

14. *Social Confinement and Mesoscopic Localization of Epidemics on Networks*
G. St-Onge, V. Thibeault, A. Allard, L. J. Dubé, L. Hébert-Dufresne 2021
Physical Review Letters. 126, 098301

13. *Inference, Model Selection, and the Combinatorics of Growing Trees*
G. T. Cantwell, **G. St-Onge**, J.-G. Young 2021
Physical Review Letters. 126, 038301

12. *Master equation analysis of mesoscopic localization in contagion dynamics on higher-order networks*
G. St-Onge, V. Thibeault, A. Allard, L. J. Dubé, L. Hébert-Dufresne 2021
Physical Review E. 103, 032301

11. *Localization, epidemic transitions, and unpredictability of multistrain epidemics with an underlying genotype network*
B. J. M. Williams, **G. St-Onge**, L. Hébert-Dufresne 2021
PLOS Computational Biology. 17, e1008606

10. *Threefold way to the dimension reduction of dynamics on networks: an application to synchronization*
V. Thibeault, **G. St-Onge**, L. J. Dubé, P. Desrosiers 2020
Physical Review Research. 2, 043215

9. *Network comparison and the within-ensemble graph distance*
H. Hartle, B. Klein, S. McCabe, A. Daniels, **G. St-Onge**, C. Murphy, L. Hébert-Dufresne 2020
Proceedings of the Royal Society A. 476, 20190744

8. *Thresholding normally distributed data creates complex networks*
G. T. Cantwell, Y. Liu, B. F. Maier, A. C. Schwarze, C. A. Serván, J. Snyder, **G. St-Onge** 2020
Physical Review E. 101, 062302

7. *Phase transition in the recoverability of network history*
J.-G. Young, **G. St-Onge**, E. Laurence, C. Murphy, L. Hébert-Dufresne, P. Desrosiers 2019
Physical Review X. 9, 041056

6. *Efficient sampling of spreading processes on complex networks using a composition and rejection algorithm*
G. St-Onge, J.-G. Young, L. Hébert-Dufresne, L. J. Dubé 2019
Computer Physics Communications. 240, 30

5. *Universality of the stochastic block model* 2018
J.-G. Young, **G. St-Onge**, P. Desrosiers, L. J. Dubé
Physical Review E. 98, 032309

4. *Phase transition of the susceptible-infected-susceptible dynamics on time-varying configuration model networks* 2018
G. St-Onge, J.-G. Young, E. Laurence, C. Murphy, L. J. Dubé
Physical Review E. 97, 022305

3. *Geometric evolution of complex networks with degree correlations* 2018
C. Murphy, A. Allard, E. Laurence, **G. St-Onge**, L. J. Dubé
Physical Review E. 97, 032309

2. *Exact vectorial model for nonparaxial focusing by arbitrary axisymmetric surfaces* 2016
D. Panneton, **G. St-Onge**, M. Piché, S. Thibault
Journal of the Optical Society of America A. 33, 801

1. *Needles of light produced with a spherical mirror* 2015
D. Panneton, **G. St-Onge**, M. Piché, S. Thibault
Optics Letters. 40, 419

Prépublications et manuscrits soumis

- *Probability generating functions for epidemiological time series*
M. C. Boudreau, **G. St-Onge**
arXiv:2610.17138.

- *The impact of behavioral homophily and conformity on epidemic spreading in networks with large groups*
O. Ribordy, C. Granell, **G. St-Onge**, L. Hébert-Dufresne, A. Arenas, A. Allard
arXiv:2605.25875.

- *Simpson's paradox explains the ubiquity of nonlinear, threshold, and complex contagions*
L. Hébert-Dufresne, A. Allard, J.-G. Young, W. H. W. Thompson, **G. St-Onge**
arXiv:2605.00791.

- *Group dynamics shape contagion onsets and multistable active phases under collective reinforcement*
S. Lamata-Otín, F. Malizia, L. A. Keating, **G. St-Onge**, V. Latora, J. Gómez-Gardeñes, L. Hébert-Dufresne
arXiv:2603.28566.

- *A multi-scale model to evaluate airport wastewater surveillance and ICU genomic monitoring for pandemic preparedness*
B. K. Reddy, J. L.-H. Tsui, K. O. Drake, **G. St-Onge**, J. T. Davis, C. Mills, J. Dunning, I. I. Bogoch, S. V. Scarpino, S. Bhatt, O. G. Pybus, A. Rambaut, M. J. Wade, T. Ward, M. Chand, E. M. Volz, A. Vespignani, M. U. G. Kraemer
medRxiv 2026.02.27.26347250.

- *Detecting structural perturbations from time series with deep learning*
E. Laurence, C. Murphy, **G. St-Onge**, X. Roy-Pomerleau, V. Thibeault
arXiv:2006.05232.

Brevets

- *Hybrid nanocomposite materials, laser scanning system and use thereof in volumetric image projection*, 2023 (délivré)
C. Allen, S. Thibault, A. Talbot-Lanciault, P. Blais, **G. St-Onge**, P. Desaulniers
Brevet CA 2983656

CONFÉRENCES ET PRÉSENTATIONS INVITÉES

- *Global Epidemic Modeling to Evaluate and Inform Wastewater Surveillance Systems* 2026
Wastewater Environmental Surveillance Meeting (présentation invitée), Ghana

- *Generating function framework for spatiotemporal computational epidemiology* 2025
Epidemics: 10th International Conference on Infectious Disease Dynamics, San Diego (CA), États-Unis

- *Future fronts: Computational Epidemiology (panel, panéliste)* 2025
Charting Complexity, Londres, Royaume-Uni

- *Model Design and Systematic Biases in Contagion Dynamics* 2025
  SIAM Conference on Applications of Dynamical Systems (DS25) (présentation invitée), minisymposium *Inference and Prediction in Complex Social Systems*, Denver (CO), États-Unis
- *Modeling Platform for Travel-Based Genomic and Wastewater Outbreak Surveillance* 2025
  - InsightNet Annual Meeting, Salt Lake City (UT), États-Unis
  - Epistorm Annual Meeting, Boston (MA), États-Unis
- *The Unreasonable Effectiveness of Branching Processes for Outbreak Analytics* 2025
  Network Science Research Symposium (présentation invitée), Boston (MA), États-Unis
- *L'efficacité hors norme des fonctions génératrices pour modéliser les épidémies* 2025
  Centre Interdisciplinaire en Modélisation Mathématique de l'Université Laval, Québec (QC), Canada
- *Statistical physics of epidemics with applications to global biosurveillance* 2025
  PHYS 7210 - Introduction to Research in Physics (séminaire), Northeastern University, Boston (MA), États-Unis
- *Generating function methodology for metapopulation epidemics with applications to global biosurveillance* 2024
  Quantitative Methods for Dynamics on Networks, Los Alamos (NM), États-Unis (présentation invitée)
- *Optimization of a global wastewater surveillance network at airports for emerging pathogens* 2024
  International School and Conference on Network Science, Québec (QC), Canada
- *Establishing a wastewater global surveillance network at airports for early detection of emerging pathogens: A modeling study* 2023
  Epidemics: 9th International Conference on Infectious Disease Dynamics, Bologne, Italie
- *Wastewater environmental Surveillance for Pandemic Preparedness (discussion de table ronde, panéliste)* 2023
  Grand Challenges Annual Meeting, Dakar, Sénégal
- *Probability generating functions for epidemics on metapopulation networks* 2023
  - Contagion on Complex Social Systems (CCSS), Burlington (VT), États-Unis
  - International School and Conference on Network Science, Vienne, Autriche
- *Quantifying population dynamics of complex contagions* 2023
  International School and Conference on Network Science, Vienne, Autriche
- *Navigating wastewater surveillance at airports with probability generating functions* 2023
  NetPLACE (présentation invitée, virtuel)
- *Indistinguishability of simple and complex contagions when transmission settings matter* 2023
  Mathematical Institute, University of Oxford, Oxford, Royaume-Uni (présentation invitée, virtuel)
- *Confounders of interacting diseases* 2023
  Dynamics of Interacting Contagions, Santa Fe (NM), États-Unis
- *Reconstruction of Product-Diffusion Cascades* 2022
  Workshop on Network Dynamics and Choice Theory, Burlington (VT), États-Unis
- *Nonlinear infection rate to compress mechanistic epidemic models* 2022
  Fifth Northeast Regional Conference on Complex Systems, Buffalo (NY), États-Unis
- *Influential groups in hypergraph contagions* 2022
  Max Planck Institute for Mathematics in the Sciences, Leipzig, Allemagne (présentation invitée, virtuel)
- *Bursty exposure on higher-order networks leads to nonlinear infection kernels* 2021
  - Networks 2021: A Joint Sunbelt and NetSci Conference, Bloomington (IN), États-Unis
  - SIAM Conference on Applications of Dynamical Systems (DS21), Portland (OR), États-Unis
  - Fourth Northeast Regional Conference on Complex Systems, Buffalo (NY), États-Unis 🏆
- *Influence maximization in simplicial contagion* 2020
  International School and Conference on Network Science, Rome, Italie
- *Localization, bistability and optimal seeding of contagions on higher-order networks* 2020
  Artificial Life Conference, Montréal (QC), Canada

- *Mesoscopic localization of spreading processes on networks* 2019
  International School and Conference on Network Science, Burlington (VT), États-Unis
- *SIS dynamics on time-varying random networks* 2017
  Institute for Disease Modeling, Seattle (WA), États-Unis
- *Susceptible-infected-susceptible dynamics on the rewired configuration model* 2017
  International School and Conference on Network Science, Indianapolis (IN), États-Unis
- *Co-evolution of Growth and Dynamics on Network* 2016
  International School and Conference on Network Science, Séoul, Corée du Sud
- *Modeling ultra-sharp needles of light using vector diffraction theory* 2014
  50th Canadian Undergraduate Physics Conference, Kingston (ON), Canada

AUTRES EXPÉRIENCES PERTINENTES

Stages

Vermont Complex Systems Center, Burlington (VT), États-Unis

- Étudiant aux cycles supérieurs en visite | groupe du Prof. Laurent Hébert-Dufresne 2019–2020
  Projet : *Reconstruction temporelle des réseaux par passage de messages*

Université Laval, Québec (QC), Canada

- Assistant de recherche de premier cycle | groupe du Prof. Louis J. Dubé 2015
  Projet : *Physique statistique des réseaux complexes*
- Assistant de recherche de premier cycle | groupe du Prof. Michel Piché 2014
  Projet : *Modélisation de faisceaux laser fortement focalisés*
- Assistant de recherche de premier cycle | groupe de la Prof. Claudine Allen 2013
  Projet : *Développement d'un système optique pour la biodétection*

Écoles et ateliers

- AI for Disease Modeling Convening, Dakar, Sénégal 2025
- Summer Institute in Statistics and Modeling in Infectious Diseases (virtuel) 2022
- Complex Systems Summer School, Santa Fe (NM), États-Unis 2018
- Complex Networks Winter Workshop, Québec (QC), Canada 2018

LEADERSHIP ET SERVICE

Conférences et ateliers

- Co-président (satellites) : International School and Conference on Network Science (NetSci2026) 2026
  - Conférence phare de la Network Science Society
- Co-organisateur : The Art of Epidemics: Data Storytelling through Effective Visualizations 2025
  - Session spéciale à Epidemics 10: 10th International Conference on Infectious Disease Dynamics
- Co-président (école et satellites) : International School and Conference on Network Science (NetSci2024) 2024
  - Conférence phare de la Network Science Society
  - Élue *Événement scientifique de l'année* par le Cercle des ambassadeurs de Québec 🏆
- Co-organisateur : Epistorm Rt-Collabathon 2024
  - Événement collaboratif autour de l'estimation en temps réel du nombre de reproduction effectif

  - Soutenu par l'initiative Insight Net du Center for Forecasting and Outbreak Analytics du CDC
- Comités de programme :
  - Conference on Complex Systems (CCS) 2024–2025
  - Northeast Regional Conference on Complex Systems (NERCCS) 2022
- Président de session :
  - Networks 2021: A Joint Sunbelt and NetSci Conference, S14 – Epidemiology 2021
  - SIAM Conference on Applications of Dynamical Systems (DS21), CP4 – Dynamics 2021
- Agent de liaison : Complex Networks Winter Workshop 2019

Comités du corps professoral

- Membre : Full Time Non-Tenure Track Faculty Committee, College of Science, Northeastern University 2024–2026
- Membre : PhD Admission Committee, Network Science, Northeastern University 2025

Évaluateur

- Revues (17) : Science Advances, Nature Communications, Physical Review Letters, Physical Review X, Physical Review E, PLOS Computational Biology, npj Complexity, PNAS Nexus, Journal of The Royal Society Interface, Journal of Complex Networks, Communications Physics, Scientific Reports, Chaos: An Interdisciplinary Journal of Nonlinear Science, New Journal of Physics, IMA Journal of Applied Mathematics, Advances in Complex Systems, PLOS One
- Triage des projets pour The Interdisciplinary Contest in Modeling (ICM) 2022

COUVERTURE MÉDIATIQUE

- À propos de notre article dans *Nature Medicine* concernant la surveillance des eaux usées des avions : 2025
  - Thinking globally for pandemic early warning systems, *Nature News and Views*
  - Wastewater sampling could be key to early warning of new disease outbreaks, *The Guardian*
  - Waste surveillance at just 20 airports could spot the next pandemic, *New Scientist*
  - How monitoring wastewater from international flights can serve as an early warning system for the next pandemic, *Northeastern Global News*
- Autre couverture médiatique de ma recherche :
  - *Mathematical model offers new insights into spread of epidemics*, *phys.org* 2021
  - *To find the right network model, compare all possible histories*, *phys.org* 2021
  - *How large a gathering is too large during the coronavirus pandemic?*, *Science News* 2020

DÉVELOPPEMENT DE LOGICIELS

- **pgfgleam** : solution efficace d'épidémies stochastiques de métapopulation pour la biosurveillance mondiale (Python)
- **SamplableSet** : implémentation d'ensembles pouvant être échantillonnés aléatoirement de manière efficace (C++/Python)
- **fasttr** : échantillonneur uniforme pour la reconstruction temporelle d'arbres croissants (C++/Python)
- **spreading_CR** : algorithme de simulation stochastique pour les processus de contagion (C++/Python)